

Module 3: Organic Molecules - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Why is carbon the backbone of most biological molecules?

- A. Carbon is the heaviest atom in cells.
- B. Carbon cannot bond to other atoms.
- C. Carbon forms up to four stable bonds, building diverse chains and rings.
- D. Carbon is found only in sugars.
- Answer: C
- Why: Carbon's ability to form four stable bonds allows large, varied molecules, which is the basis of life's chemical diversity.

2. Cells join two monomers into a polymer by removing a water molecule. This reaction is called:

- A. dehydration synthesis
- B. hydrolysis
- C. denaturation
- D. diffusion
- Answer: A
- Why: Dehydration synthesis links monomers by removing water. Hydrolysis is the reverse: adding water to break the bond.

3. A protein is heated until it loses its three-dimensional shape and stops working. This change is called:

- A. replication
- B. polymerization
- C. hydrolysis
- D. denaturation
- Answer: D

- Why: Denaturation is loss of shape. Because a protein's function depends on its shape, the protein stops working.

4. Which pairing of molecule and job is correct?

- A. Lipids store genetic information.
- B. Nucleic acids store genetic information.
- C. Carbohydrates always speed up reactions.
- D. Proteins are the main long-term energy store.
- Answer: B
- Why: Nucleic acids (DNA and RNA) store information. Carbohydrates and lipids store energy, and proteins do most cellular work.